



Gesture-controlled musical interface enabled mechano-electric transducer using intrinsic hybrid nanogenerator[☆]

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ABSTRACT

Gesture-controlled musical interfaces (GCMI) offer an innovative approach to interacting with sound through body movements, enabled by mechano-electric transducers composed of an intrinsic hybrid nanogenerator (IHNG). The IHNG was constructed through 0.3Ba_{0.7}Ca_{0.3}TiO₃–0.7BaSn_{0.12}Ti_{0.88}O₃ (BCST)/polydimethylsiloxane (PDMS) composite film. Enhanced piezoelectric ($d_{33} = 205$ pC/N) and ferroelectric ($P_r = 8.06$ μC/cm², $E_c = 3.4$ kV/cm) properties of BCST ceramic were observed by optimizing the mechanical load (10–50 kN), resulting in the generation of the nanopolar regions, rhombohedral–orthorhombic–tetragonal phase coexistence, and switchable domains. The IHNG device generates an output of 250 V and 2.4 mA/m² under mechanical load, driven by the synergistic effects of piezoelectric and triboelectric mechanisms. Furthermore, the GCMI system realised through the IHNG-based smart glove functioned as a mechano-electric transducer to demonstrate the real-time air drum.

1. Introduction

Gesture-controlled musical interfaces (GCMI) have gained significant interest due to easy operation through gesture movements, cost-effectiveness, and wide utility in various fields, including easy access for persons with disabilities. The GCMI allows the user to interact with the hardware unit through gestures, enabling control and instruction of musical sounds using a programmed set of conditions to execute the necessary tasks, or to request help from a third party. These interfaces rely on mechano-electric transducers that can be developed using haptic sensors, gesture-controlled sensors, and MIDI controllers, eliminating the need to press buttons and turn knobs. Recently, triboelectric nanogenerators (TENG) and piezoelectric nanogenerators (PENG) are widely explored to convert waste mechanical energy to electrical energy due to their easy fabrication and cost-effective nature [1].

TENG can be used for mechano-electric transducers by utilizing coupled effect of contact electrification and electrostatic induction [2]. It can be used for various applications like haptic sensing [3], AR/VR technology [4], wearables [5], motion monitor [6], tactile [7], gesture recognition [8], healthcare systems [9], and rehabilitation [10]. Piezoelectric nanogenerators (PENG) are energy harvesters that convert mechanical stimuli into electrical signals. PENG was used as the mechano-electric transducer in applications like self-powered electronics [11], pressure sensor [12], biomedical devices [13] and wound healing [14]. Hybridizing TENG and PENG creates a reliable and efficient mechano-electric transducer through a synergistic effect, resulting in enhanced sensing behaviour and accuracy. Yuan et al. reported the piezo-tribo hybrid nanogenerator using the extrinsic hybridization with two standalone harvesters [15]. Similarly, Prasanna et al. prepared the extrinsic piezo-tribo hybrid nanogenerator by the integrated

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configuration of the two harvesters [16]. But the extrinsic hybridization requires additional circuit management for coupling the signals of different harvesters, which can create signal mismatch and electric noise from the surrounding environment. To overcome the issue, Monunith et al. reported the intrinsic hybridisation-based two-electrode piezo-tribo hybrid nanogenerator with an internally connected multilayered composite film for the simultaneous piezoelectrification and triboelectrification [17]. This approach will reduce the number of device layers, fabrication time, and eliminate the need for additional circuit components. But the configuration is limited to the single electrode mode TENG-based harvesting and sensing. The limitation is resolved by a multi-layered multifunctional composite film-based three-electrode configured piezo-tribo intrinsic hybrid nanogenerator. To prepare the multi-layered multifunctional composite film, optimization of piezoelectric and triboelectric materials is required.

Compositional modifications through strategic doping at the A-site, B-site, or both sites, processing temperatures, soaking times, and electrical poling techniques were extensively studied in order to enhance the piezoelectric and ferroelectric properties of ABO₃-type perovskite ceramics. The lead-free ceramics have a higher longitudinal piezoelectric coefficient ($d_{33} = 100$ pC/N to 530 pC/N) than the piezoelectric polymers (PVDF and its copolymers = ≤ -34 pC/N), but lower than the lead-based piezoelectrics such as PbZrTiO₃, PbMgNbO₃-PbTiO₃, and so on. The use of lead piezoelectrics is being restricted globally due to concerns about toxicity, environmental pollution, and their potential to cause cancer and nerve-related health disorders. Therefore, unlocking the piezoelectric properties of lesser-explored lead-free ceramics has sparked global attention, as they have the potential to replace lead-based ceramics in all major fields. Among these, BaTiO₃ (BT, $d_{33} \approx 100$ pC/N) is one such biocompatible ceramic whose functional properties were tailored by doping, growth process, operating temperatures, texture orientation, and crystallinity. Elorika et al. reported an increase in the d_{33} value from 97 to 110 pC/N for the polycrystalline Ba_{1-x}Ca_xTiO₃ composition ($x = 0, 0.03, 0.08$) due to the substitution of Ca²⁺ ions at the A-site (Ba²⁺ ions) of BT [18]. They also reported a decrease in the d_{33} value of up to 87 pC/N when the concentration of $x = 0.05$, and noticed that the partial substitution of Ca²⁺ ions for Ti⁴⁺ ions creates disorder and defects in the crystal lattice, resulting in the decreased d_{33} value. Nawal et al. studied the structural, dielectric, and piezoelectric properties of polycrystalline BaZr_xTi_{1-x}O₃ ($x = 0, 0.05, 0.10, 0.15, 0.20$) composition powders (i.e. partial substitution of Zr⁴⁺ ions for Ti⁴⁺ ions) synthesized by sol-gel approach [19]. A higher d_{33} value of 126 ± 5 pC/N was observed for the BaZr_{0.05}Ti_{0.95}O₃ composition, which has an orthorhombic structure and good densification compared to all other compositions. Monunith et al. reported simultaneous A-site and B-site doping of the BaTiO₃ lattice (i.e. 0.3Ba_{0.7}Ca_{0.3}TiO₃-0.7BaSn_{0.12}Ti_{0.88}O₃), as well as the introduction of additional Cu₂O material to enhance sinterability during a solid-state reaction [17]. The material exhibited a d_{33} value of 108 pC/N and stable ferroelectric properties with minimal leakage loss. Here, they investigated the role of sintering temperature (1350 °C to 1550 °C for 6 h) and the concentration of Cu₂O on the functional properties of ceramics, resulting in a slight improvement in the properties. There is a possibility to achieve a higher d_{33} value in the above-mentioned ceramic without the use of Cu₂O and higher sintering temperatures. This can be achieved by increasing the density of the pellet, formation of nanopolar regions, promoting the coexistence of multiphases, and ensuring the phase stability of the pellet [20]. It can be achieved through the systematic study of various pelletization loads (10 kN to 50 kN) of the ceramic at specific, suitable sintering temperatures and soaking times. Overall, achieving high piezoelectric performance of lead-free fillers and creating efficient composite films that are compatible for piezoelectric, triboelectric, or synergistic effects is highly necessary for next-generation harvesters, sensors, and specifically for GCMI.

Here, a binary composition nanopowder, i.e. 0.3Ba_{0.7}Ca_{0.3}TiO₃-0.7BaSn_{0.12}Ti_{0.88}O₃ (BCST) is synthesized by solid-state reaction,

and its structural, piezoelectric, and ferroelectric properties were measured by fabricating ceramic pellets through various pelletization loads (10 kN to 50 kN) at specific, suitable sintering temperatures and soaking times. At a 30 kN load, the pellet exhibits a higher d_{33} value of 205 pC/N, reasonable remnant polarization of 8.06 $\mu\text{C}/\text{cm}^2$, and minimized losses. This is attributed to the formation of nanopolar regions and the existence of a morphotropic phase boundary (MPB) or a near MPB characterized by the coexistence of rhombohedral (R), orthorhombic (O), and tetragonal (T) phases. The flexible multifunctional composite film is fabricated by dispersing of 20 wt% of piezoelectric BCST filler particles (obtained from a 30 kN load pellet) in a triboactive polydimethylsiloxane (PDMS) polymer [21]. The BCST/PDMS composite film can exhibit both piezoelectric and triboelectric effects with enhanced surface charge density and dielectric properties. Leveraging these enhanced properties, an intrinsic piezoelectric-triboelectric hybrid nanogenerator (IHNG) was developed to achieve a synergistic energy harvesting effect. The IHNG, featuring multiple electrodes, was configured into two distinct setups, designated IHNG-1 and IHNG-2. Both configurations achieve superior electrical outputs compared to PENG, single-electrode TENG (SE-TENG), and contact-separation TENG (CS-TENG). Following optimization, the IHNG was successfully utilized as a mechano-electric transducer. Furthermore, the GCIM system is designed to showcase the different air drum functionalities using gesture movements (i.e. Human-machine interface) through a network of mechano-electric transducers, signal processing unit, software analyzer, and display units.

2. Results and discussion

Fig. 1a shows the schematic illustration of the material preparation and device fabrication of the intrinsic hybrid nanogenerator. In the experimental section, detailed procedures for material preparation and device fabrication were provided. Fig. 1b shows the layer-by-layer view of the intrinsic hybrid nanogenerator and digital image of the gesture-controlled musical interface smart glove. Gesture-controlled musical interface was constructed with series networks using the smart glove (prepared using mechano-electric transducer), signal processing unit, signal analyzer, and display and sound system, as shown in Fig. 1c. The mechano-electric transducer (intrinsic hybrid nanogenerator) converts the mechanical trigger into electrical signals. The signal processing unit constructed with rectifier unit and a signal conditioning unit. The rectifier unit converts the generated AC electrical signals into DC signals. The signal conditioning unit converts the electrical signals into Arduino analog read values. The signal conditioning unit was constructed by using the Arduino Nano 33 BLE. The Arduino analog read values were analysed by the developed code using software named Processing IDE. Once the signal reaches the threshold limit, the software produces the drum's respective sound with mechanical triggers. In this work, the threshold limit is set to be 800 analog read values. Before constructing the GCMI system, the material, device, and mechano-electric sensor were characterized.

BCST is a multi-crystalline material, and its phase compositions were altered with the influence of sintering temperature (1450 °C) and applied mechanical load (10 kN to 50 kN) during the sintering. Fig. 2a shows the XRD pattern of the BCST materials prepared at different applied load. XRD pattern shows the perovskite structure with the multi-crystalline phase existence [17]. The peaks around 39°, 45° and 56° are considered the key characteristic peaks to observe the change in the phases of the perovskite structure [22,23]. The peak splitting around 45° and 56° confirms the presence of tetragonal and orthorhombic phases coexistence [16,24]. For depth understanding of crystal structure and lattice symmetry, Rietveld refinement analysis was carried out using the FullProf software. Fig. 2b and Fig. S1 show the Rietveld refinement fit of the different load applied BCST samples. The refinement analysis reveals the composition of P4mm (tetragonal), Amm2 (orthorhombic), and R3m (rhombohedral) phase coexistence in the different load applied BCST

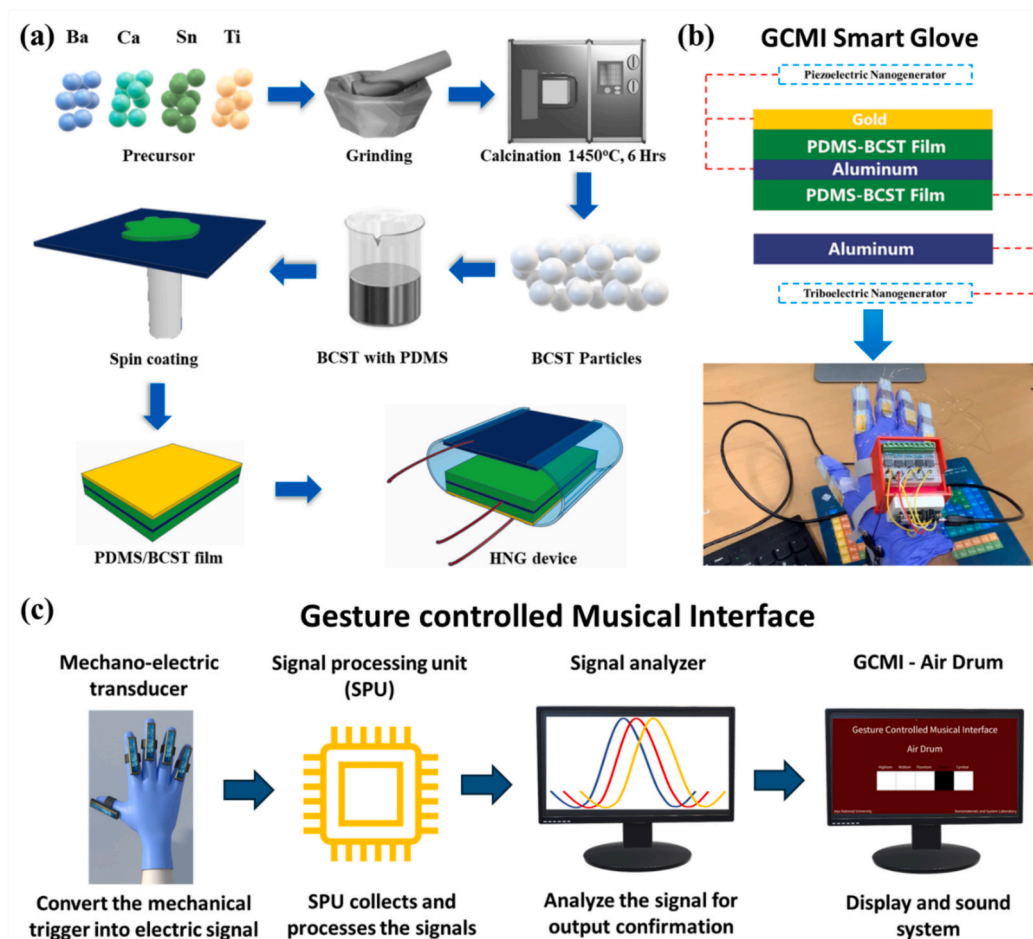


Fig. 1. (a) Schematic illustration of the material and device preparation, (b) layer-by-layer scheme of the intrinsic hybrid nanogenerator and digital image of the GCMI smart glove, and (c) working process of the GCMI system.

samples. Fig. 2c shows the phase analysis comparative plot of the different BCST samples. In general, the existence of the polar phases (tetragonal, orthorhombic, and rhombohedral) creates the net dipole moment, which is useful for charge generation under mechanical stress. With the applied mechanical load, different BCST samples show a change in the phase composition. 30 kN load applied BCST sample was found to be the tipping point for the best composition of the R-O-T phase coexistence. The quantified values of atomic position, percentage of occupancies, and lattice parameters were evaluated using the Rietveld refinement analysis and tabulated in Tables S1-S5 for different BCST samples. To understand the temperature influence in the phase change, in-situ high-temperature XRD measurements were carried out, as shown in Fig. 2d. Fig. 2e and f shows the contour plot of (111) and (002/200) plane of the perovskite structure. The plot clearly shows that there is a decrease in the angle of the diffraction and intensity with the increase of temperature for the (111) and (002/200) planes. Shifting towards the lower angle indicates the expansion of the lattice with the temperature [25]. Fig. S2 shows the magnified view of the (111) plane and (002/200) plane XRD pattern of all different mechanical loads applied BCST samples. The XRD pattern clearly shows the change in the diffraction angle and intensity. Any change or modulation in (111) plane results in the octahedral tilt, and a change in the (002/200) plane causes the lattice distortion [26,27]. According to Bragg's law, the diffraction pattern depends on $2d \sin \theta = n\lambda$ (where n is the order of diffraction, λ is the wavelength of the incident x-ray, θ is the angle of diffraction, and d is the interplanar spacing) [28,29]. In general, for a fixed wavelength, the diffraction angle is inversely proportional to the interplanar spacing.

The change in the diffraction angle with all the results suggests the change in the interplanar distance and crystal distortion to create the phase transition due to the thermal influence and applied mechanical load.

Using the Raman spectroscopy multiphase coexistence, dynamic symmetry and lattice vibration were evaluated for different load applied BCST samples (Fig. 2g). In general, the region 100 cm^{-1} to 300 cm^{-1} in Raman spectra are affected by the polymorphic transitions. And these regions are mainly related to polar BO_6 vibrations associated with the multiphases. The peak between 185 cm^{-1} to 192 cm^{-1} are associated with longitudinal optical phonon modes of rhombohedral or orthorhombic phases [30,31]. The broad peak from 210 cm^{-1} to 260 cm^{-1} reveals the phase transition of tetragonal and orthorhombic phases [32]. The peak around 300 cm^{-1} is due to the longitudinal and transverse vibration associated with the tetragonal phases [16,26]. Fig. S3 and Fig. 2h show the FTIR position mapping of the 30kN load applied BCST sample overall mapping and different vibrations associated with the metal-oxide bonds (Ba—O, Ti—O and Ti—O—Ti) of the perovskite structures. The colour variation in the mapped image shows the change in intensity of a particular bond vibration in the mapped area. The intensity of IR peak was altered by the local chemical environment or change in dipole moment during vibration. The local environment can be affected due to the dopant chemical composition used in the sample results in the change of bond length between metal-oxides. The difference in bond length creates the change in dipole moment ($\partial\mu/\partial x$) during the vibration. It clearly suggests the different phase boundaries exist in the sample. XRD, FTIR and Raman spectra results shows the coexistence

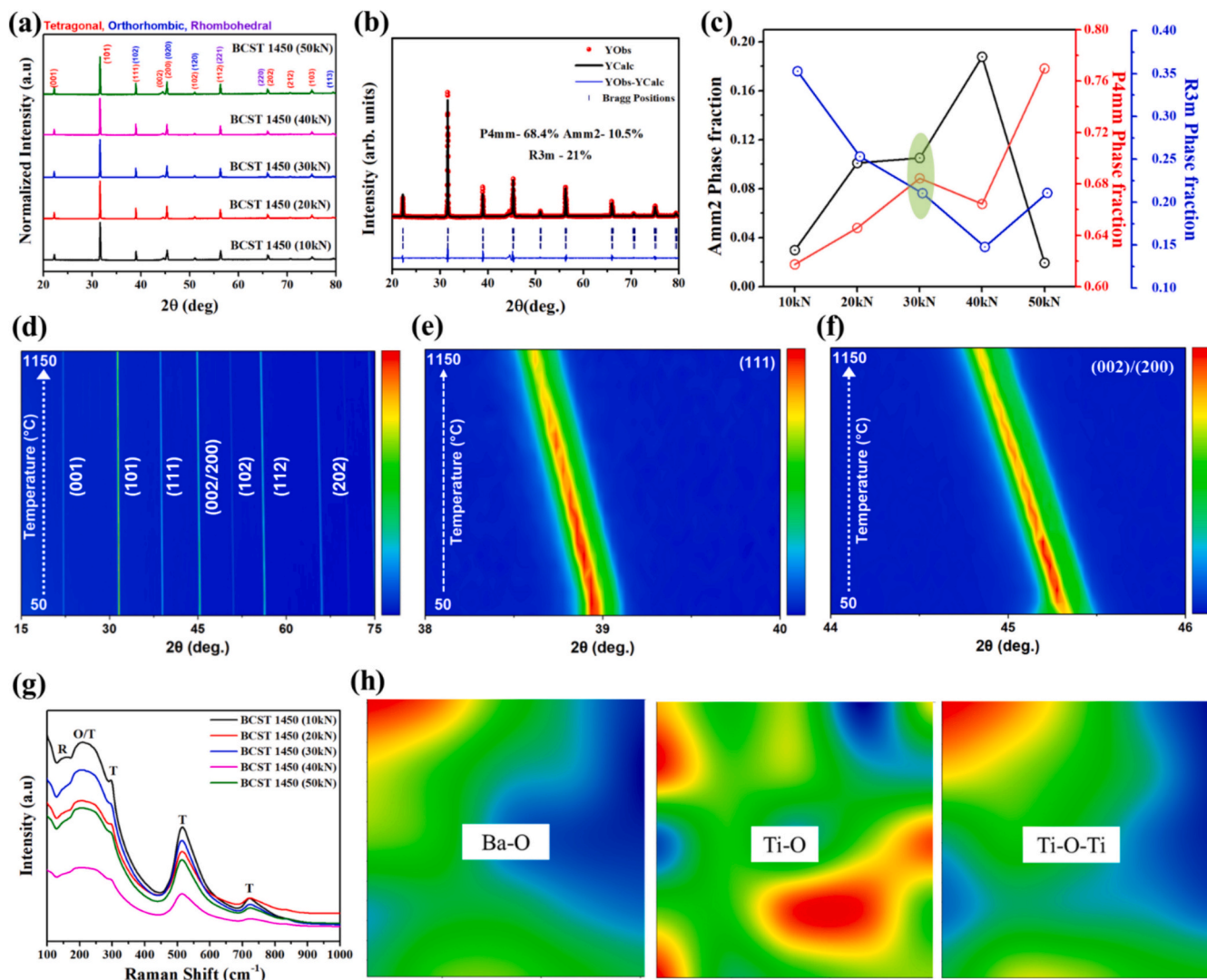


Fig. 2. (a) XRD pattern of different load applied BCST samples, (b) Rietveld refinement fit of 30 kN load applied BCST sample, (c) comparative plot of R-O-T phase percentage in different load applied BCST samples, high-temperature in-situ XRD analysis of 30 kN load applied BCST sample, (d) overall XRD pattern, (e) (111) plane in-situ XRD analysis, (f) (002)/(200) planes in-situ XRD analysis, (g) Raman spectrum of different load applied BCST samples, and (h) FTIR mapping of different metal oxides bonds in perovskite structure.

of the multiphases with the possibilities of MPB or near MPB. Fig. S4 depicts the SEM image of different load applied BCST samples. All the samples show the well-densified coarse-grained irregular grains. The pores were reduced in the microstructure with the increased applied mechanical load to the BCST samples. Fig. S5 shows the elemental mapping of the 30 kN load applied to the BCST sample. The elements barium, calcium, tin, titanium, and oxygen were present in the sample.

After the materials characterization of different load applied BCST pellets, the ferroelectric, piezoelectric and dielectric studies were carried out at room temperature. Fig. 3a depicts the polarization-electric field (P-E) hysteresis loop for the different load applied BCST samples. The shape of loop shows the ferroelectric behaviour of different BCST sample. Fig. 3b shows the comparative plot for the remnant polarization and relative difference between the remnant polarization and saturation polarization ($P_s - P_r$) for the different BCST samples. 30 kN load applied BCST sample shows the maximum remnant polarization with minimum relative difference $P_s - P_r$. This behaviour indicates the possibility of existence of the switchable domains exist in the prepared material [26]. Moreover, the increase in remnant polarization shows significant number of electric dipoles are oriented in the particular direction. Apart from that, all loop shows the low coercive field even at the applied high field

(Fig. S6a). The low value of coercive field indicates the soft ferroelectric nature of the samples. This behaviour is due to the R-O-T multiphases coexistence [32,33]. Fig. 3c shows the P-E loop of 30 kN load applied BCST sample at different applied electric field. Fig. S6b shows the comparative plot for the remnant polarization and relative difference between the remnant polarization and saturation polarization ($P_s - P_r$) for the 30 kN load applied BCST sample. The remnant polarization increases with increase in the applied field. And relative difference between the remnant polarization and saturation polarization is less than 57%. Fig. 3d shows the P-E loop and I-E (current-electric field) curve for the 30 kN load applied BCST sample at the applied electric field of 40 kV. I-E indicates two signals namely: domain switch current and leakage current. The switchable domain switch current was observed at the field of coercive field, which indicates the orientation change of electric dipoles with the positive and negative applied field [16]. Usually, leakage current was observed at the higher applied field. 30 kN load applied samples shows the low current indicate good nature of the sample [34]. To further confirm the switchable domain existence and polar nano region, PFM measurements were carried out. Domain region of $100 \text{ nm} \times 100 \text{ nm}$ was selected for the PFM analysis. The positive (10 V) and negative (−10 V) bias were applied to the region. Fig. 3e and f shows the

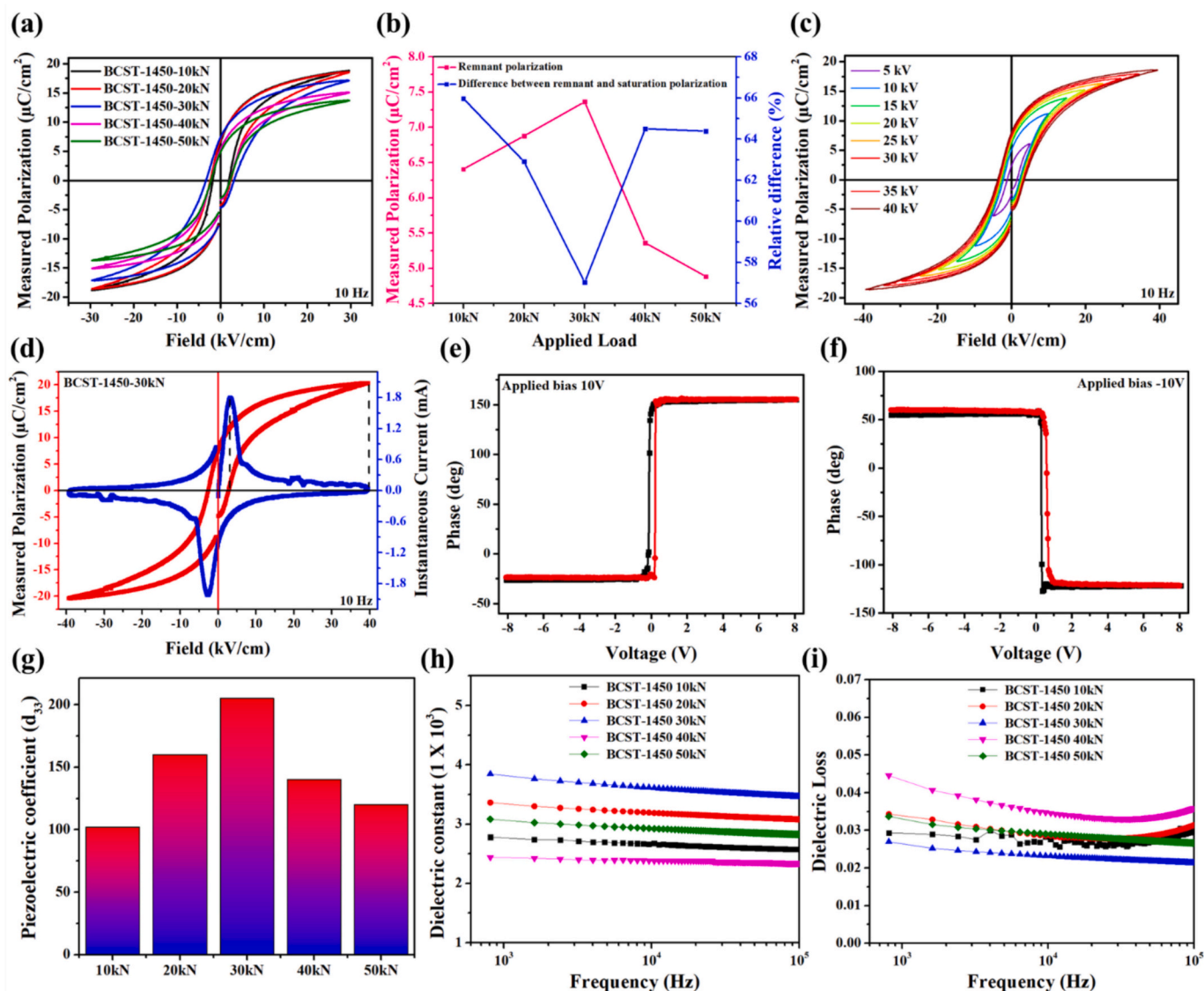


Fig. 3. (a) P-E loop analysis of different BCST samples, (b) remnant polarization and relative difference of different load applied BCST samples, (c) P-E loop analysis of BCST-1450-30kN with different applied field, (d) P-E loop and I-E curve of BCST-1450-30kN, (e-f) PFM phase loop measurement at different applied potentials (+10 V and -10 V) of BCST-1450-30kN sample, (g) piezoelectric coefficient of different load applied BCST samples, (h) dielectric constant of different load applied BCST samples, (i) dielectric loss of different load applied BCST samples.

switch in the phase loop with the different applied bias confirms the presence of switchable dipoles. Fig. S7 shows the PFM amplitude image for the applied positive (10 V) and negative (-10 V) bias. The image clearly shows there is a change in the amplitude with the change in the applied bias. Fig. 3g represents the piezoelectric coefficient (d_{33}) for the different load applied BCST samples. 30 kN load applied BCST sample shows highest d_{33} value of 205 pC/N. The dielectric constant and dielectric loss of the different load applied BCST samples are shown in Fig. 3h and i respectively. 30 kN load applied samples shows the high dielectric constant and low dielectric loss comparing with other prepared samples. Fig. S8 shows the temperature dependent dielectric studies of 30 kN load applied BCST sample. The Curie temperature (T_c) was observed near 135 °C at 1 kHz to the applied frequency for BCST-1450-30kN sample. All these results indicate improved performance, and sensitivity can be possible for the mechano-electric transducer with the optimized material.

30 kN load applied BCST sample were found to be the best sample with the phase analysis, ferroelectric, piezoelectric and dielectric property. The PDMS/BCST composite film were prepared using 30 kN

load applied BCST. Fig. 4a shows the XRD pattern of the pure PDMS and PDMS/BCST composite film. The hump around 10° - 15° indicates the amorphous nature of the pure PDMS film. After 20° crystalline peaks present in the composite film indicates the presence of the BCST in the polymer. Fig. S9 shows Raman analysis of the pure PDMS and composite film. The dominant bands of the PDMS due to the stretching of CH_3 were observed at 2700-3000 cm^{-1} . The local phonon mode of BCST were present at 150-800 cm^{-1} . Both the bands of PDMS and BCST were present in the Raman spectra of the composite films. Fig. S10 shows the SEM image and elemental mapping of pure PDMS film. The SEM image shows the smooth surface and elemental mapping confirms the presence of silicon, carbon and oxygen. Fig. S11 show the SEM image and elemental mapping of PDMS/BCST composite film. The BCST particles are evenly distributed in the polymer matrix of PDMS. The elemental mapping shows the presence of silicon, oxygen, and carbon associated with PDMS and barium, calcium, tin, and titanium due to the BCST particles. Fig. S12 shows the cross-sectional SEM image and elemental mapping of the PDMS/BCST composite film. The SEM image and mapping clearly show that the aluminium electrode is attached between two

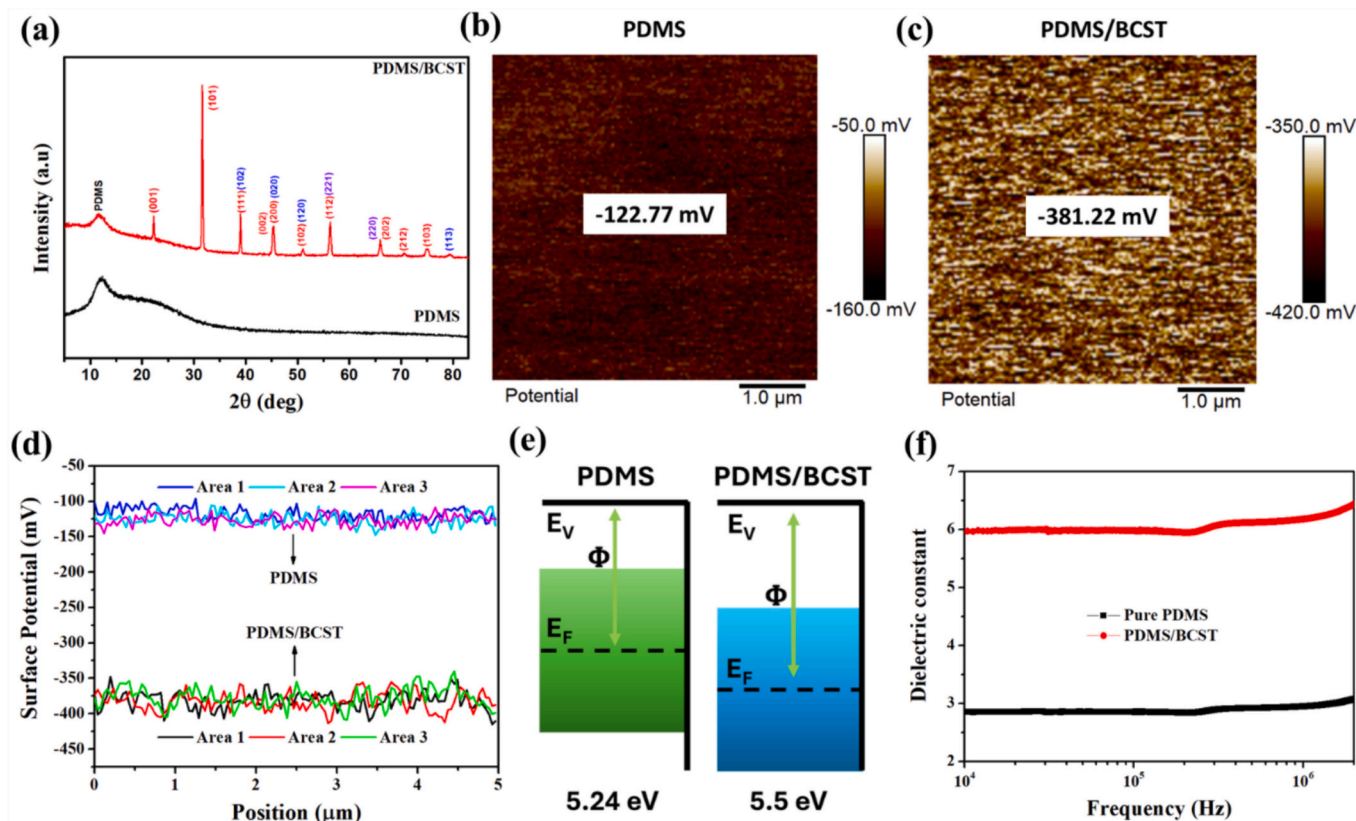


Fig. 4. (a) XRD analysis of pure PDMS and PDMS/BCST composite film, (b-c) surface potential mapping of pure PDMS and PDMS/BCST composite film, (d) horizontal line profile mapping of surface potential on pure PDMS and PDMS/BCST composite film, (e) energy band diagram of pure PDMS and PDMS/BCST composite film, and (f) dielectric constant measurement of pure PDMS and PDMS/BCST film.

PDMS/BCST composite films.

Using the PDMS/BCST composite film, all the nanogenerators, including PENG, TENG, and HNG, were fabricated. Fig. S13 a and b shows the voltage and current density comparative plot of pure PDMS and different weight ratio-based PDMS/BCST composite film TENG devices measured in the contact-separation mode. 20 wt% based PDMS/BCST composite film TENG device showed a maximum output performance (191 V and 1.38 mA/m²). The usage of the filler enhanced the output performance of a 20 wt% PDMS/BCST composite film-based TENG device by two times compared to the pure PDMS TENG device. In general, TENG device output performance depends on the work function, surface charge density, and dielectric property of the tribolayers [35,36]. Fig. 4b and c shows the surface potential mapping of the pure PDMS and PDMS/BCST films. The PDMS/BCST film has high surface potential compared with the pure PDMS film. To confirm the uniform surface potential nature of the film, horizontal line profile mapping was carried out for a distance of 5 μ m. The PDMS/BCST composite film shows the uniform surface potential comparatively higher to the pure PDMS film, as shown in Fig. 4d. The contact potential difference between the tip and sample are calculated with the following equation [36,37].

$$V_{CPD} = \frac{\Phi_{tip} - \Phi_{sample}}{e}$$

where Φ_{tip} and Φ_{sample} is the work function of the tip and sample, and e is the charge of the electron. Using the reference material, the work function of the tip is calculated, which is found to be 5.121 eV. The work function of the PDMS/BCST composite film is high compared to the pure PDMS, which indicates the high electrophilic nature of the film (Fig. 4e). Fig. 4f shows the dielectric constant measurements of pure PDMS and PDMS/BCST composite film, and it is found that the composite film

shows a high dielectric constant. The increase in the dielectric constant increases the electrostatic induction effect between the tribolayers and electrode. The results of the surface potential, work function and dielectric constant confirm that the BCST filler changes the film property, which is the key factor for the enhancement in the output performance. Fig. S14a shows the PDMS/BCST composite film-based TENG device output performance analysis with the different relative humidity (30% to 80%). The PDMS/BCST composite film-based TENG device output performance decreases with increased relative humidity. To resolve the issue, packed TENG techniques were developed [38]. Fig. S14b shows the temperature-dependent output performance analysis of the PDMS/BCST composite film-based TENG device. With the different temperatures (25 $^{\circ}$ C to 40 $^{\circ}$ C), the PDMS/BCST composite film-based TENG device shows no drastic changes or deformation in the waveform, with the output performance. ZL Wang and co-workers, with their study on the temperature effect on the TENG, claimed that between 25 $^{\circ}$ C and 60 $^{\circ}$ C, no significant change in the output performance were observable [39]. Fig. S15 a and b show the voltage and current density analysis of different weight ratio-based PDMS/BCST composite film PENG devices. 20 wt% based PDMS/BCST composite film PENG device obtained a maximum output voltage of 29 V and 0.088 mA/m².

The prepared hybrid nanogenerator contains three active layers and electrodes (two aluminium (named as Al-1 and Al-2) and one gold electrode), as shown in Fig. S16. To obtain better mechano-electric response, it is necessary to study the connections of different electrodes. In the first electrode configuration, the top gold electrode and intermediate aluminium electrode (Al-1) are connected, as shown in Fig. 5a. The other aluminium electrode Al-2 is grounded. This configuration is named as IHNG-1 (intrinsic hybrid nanogenerator-1). The configuration looks like a piezoelectric nanogenerator intrinsically connected with the single electrode mode TENG (SE-TENG). Fig. 5b and

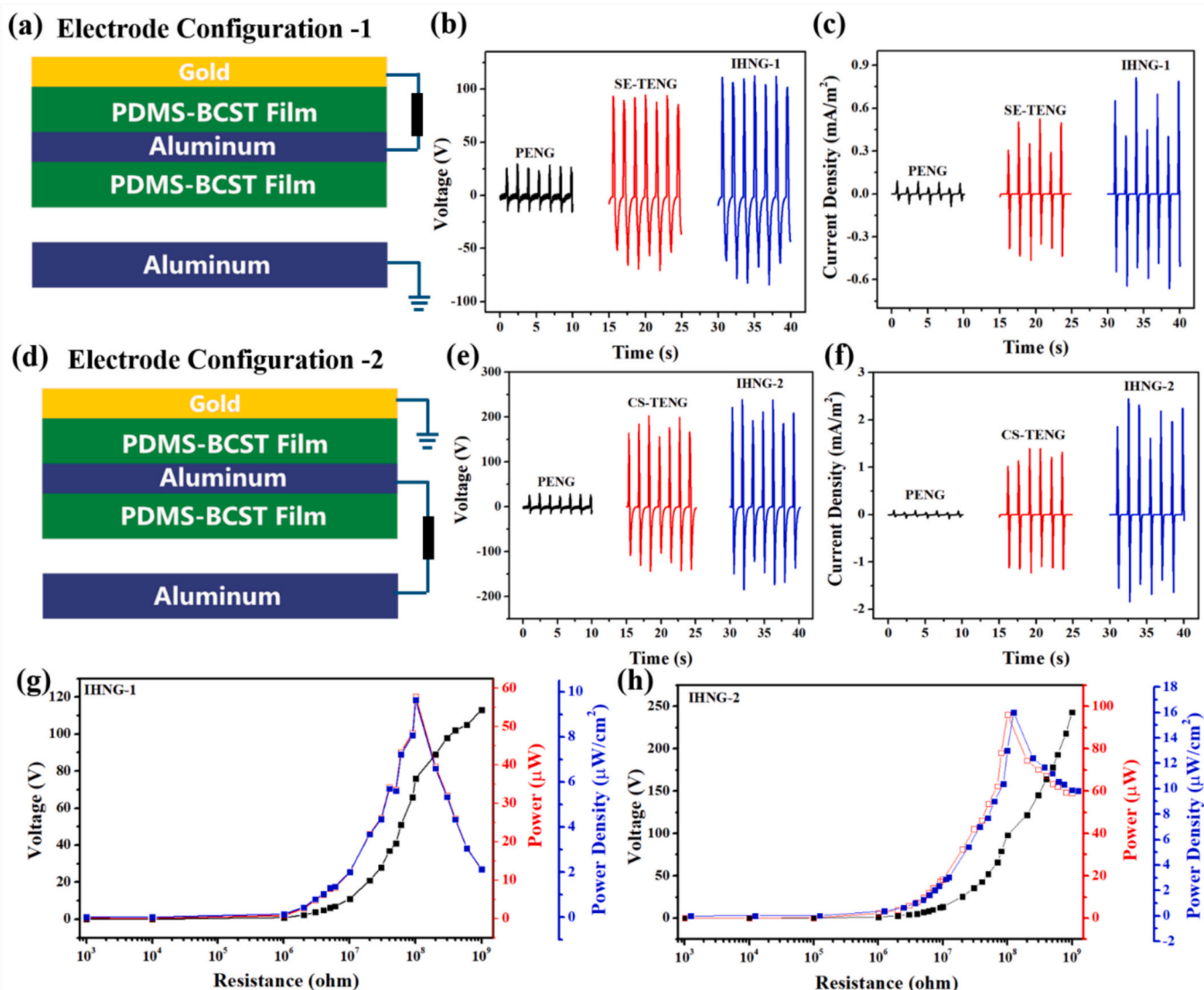


Fig. 5. (a) Schematic illustration of IHNG-1 device and electrical connection, (b) voltage comparison plot of PENG, SE-TENG and IHNG-1, (c) current comparison plot of PENG, SE-TENG and IHNG-1, (d) schematic illustration of IHNG-2 device and electrical connection, (e) voltage comparison plot of PENG, CS-TENG and IHNG-2, (f) current comparison plot of PENG, CS-TENG and IHNG-2, and (g-h) load matching analysis of IHNG-1 and IHNG-2.

c show the output voltage and current density comparison plot of PENG, SE-TENG, and IHNG-1. Compared with PENG and SE-TENG, IHNG-1 shows the maximum output voltage of 114 V and current density of 0.81 mA/m². In the second electrode configuration, both the aluminium electrodes Al-1 and Al-2 were connected for the measurements, as shown in Fig. 5d. The gold electrode is connected to the ground. This configuration is named as IHNG-2 (intrinsic hybrid nanogenerator-2). The configuration resembles the contact separation TENG (CS-TENG), intrinsically connected with the PENG device. Fig. 5e and f show the output voltage and current density comparison of PENG, CS-TENG, and IHNG-2. Compared with PENG and CS-TENG, IHNG-2 shows the maximum output voltage of 250 V and current density of 2.4 mA/m². The improved output performance is due to the synergy between piezo and tribo charges on the intermediate aluminium electrode (Al-1). Fig. 5g and h show the load matching analysis of the hybrid nanogenerator devices IHNG-1 and IHNG-2. The IHNG-1 device shows the maximum power of 57 μW and power density of 9.6 μW/cm² at the load resistance of 100 MΩ. The IHNG-2 device shows the maximum power of 96 μW and power density of 16 μW/cm² at the load resistance of 100 MΩ. Fig. 6a and b shows the stability of the hybrid nanogenerator devices IHNG-1 and IHNG-2. Both electrode configurations show stable

output generation under continuous mechanical stress for over 1000s. To ensure the capability of the prepared hybrid nanogenerator device, a commercial capacitor (0.47 μF) was charged for 300 s with the harvested power. Fig. 6c and d shows the comparative capacitor charging plot of the different prepared nanogenerator devices. Fig. 6e and f shows the comparative plot of the charge stored in the capacitor with the harvested energy from different prepared devices. The IHNG-1 device shows the maximum charged voltage and charge compared with the PENG and SE-TENG devices. Similarly, the IHNG-2 device shows the maximum charged voltage and charge compared with the PENG and CS-TENG devices. Furthermore, the harvested energy from the hybrid nanogenerator was utilized to power the low-power electronics. Fig. 6g and h shows the powering of the hygrometer and calculator from the harvested energy.

After careful investigation of the hybrid nanogenerator electrical response, it is used as a mechano-electric transducer for sensing purposes in the smart glove. The smart glove is prepared using multiple mechano-electric transducers. Each mechano-electric transducer signifies a different sound of the drum as shown in Fig. 7a. To ensure the reliability of the sensor, single sensor response, different subject studies, durability test, and consecutive response were studied. Fig. 7b shows the

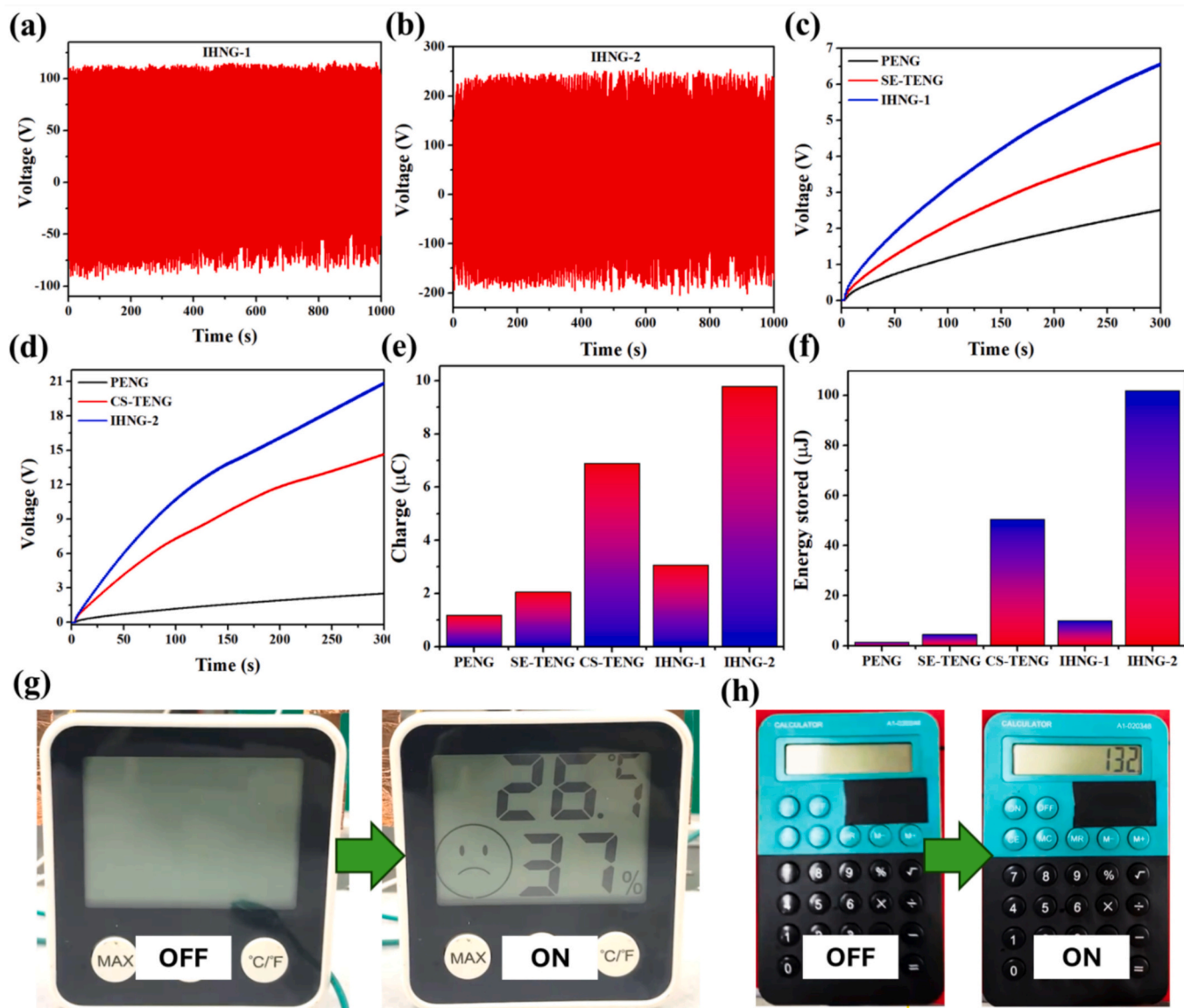


Fig. 6. Stability test of (a) IHNG-1 device, (b) IHNG-2 device, (c) capacitor charging comparison plot of PENG, SE-TENG and IHNG-1, (d) capacitor charging comparison plot of PENG, CS-TENG and IHNG-2, (e) charge stored in the capacitor of different NG-devices for the period of 300 s, (f) energy stored in the capacitor of different NG-devices for the period of 300 s, and (g-h) low power electronics powered using harvested energy of IHNG-2 device.

single sensor continuous response and all the bending of flexions were above the threshold limit. Fig. 7c shows the interference study of the multi-finger response. The results suggest that only the full flexion finger has crossed the threshold limit. Fig. 7d shows the different subject responses with the full flexion for the selection of mid tom. Fig. 7e shows the error bar graph of the different subjects for the flexions for the single selection. Fig. S17 shows the response and recovery plot of the mehanoelectric transducer. The device shows the response time of 0.1 s and the recovery time of 0.3 s. All the studies show the proper selection of the desired flexion for the respective sound. All subjects were informed about the nature of the experiment and the intended research use of their data from their participation. Furthermore, participant details were kept confidential. Fig. S18 shows the schematic diagram of the mehanoelectric transducer and processing unit. Fig. 7f shows the digital image of the GCMI system. The five-finger sensor response of the mid tom selection was shown in Fig. 7g. The results show that the sensor for the mid tom only reached the threshold limit, and the analyzer selected the mid tom sound to display. Fig. 7h continuous multi-sensor response for the continuous multiple selection. GCMI system was

demonstrated in the Supplementary Movie S1.

3. Conclusion

Gesture-controlled musical interface system was prepared using the mechano-electric transducer-based smart glove. The mechano-electric response of the transducers was optimized using the triboelectric and piezoelectric properties. The mechanical load optimization (10–50 kN) of BCST sample leads to R–O–T phase coexistence for improved piezoelectric (205 pC/N) and ferroelectric properties ($\text{Pr} = 8.06 \mu\text{C}/\text{cm}^2$). The PDMS/BCST composite film preparation improved the surface potential (-381.22 mV), work function (5.5 eV), and dielectric property (2 times higher to bare PDMS) for better triboelectrification. An intrinsic piezo-tribo hybrid nanogenerator (IHNG) was successfully fabricated using a PDMS/BCST composite film. The device comprised a double-layered PDMS/BCST film that creates two distinct electrical configurations, IHNG-1 and IHNG-2. Notably, the IHNG-2 configuration yielded the highest output, reaching a peak voltage of 250 V and a maximum power of 96 μW . The developed IHNG was used as a mechano-electric

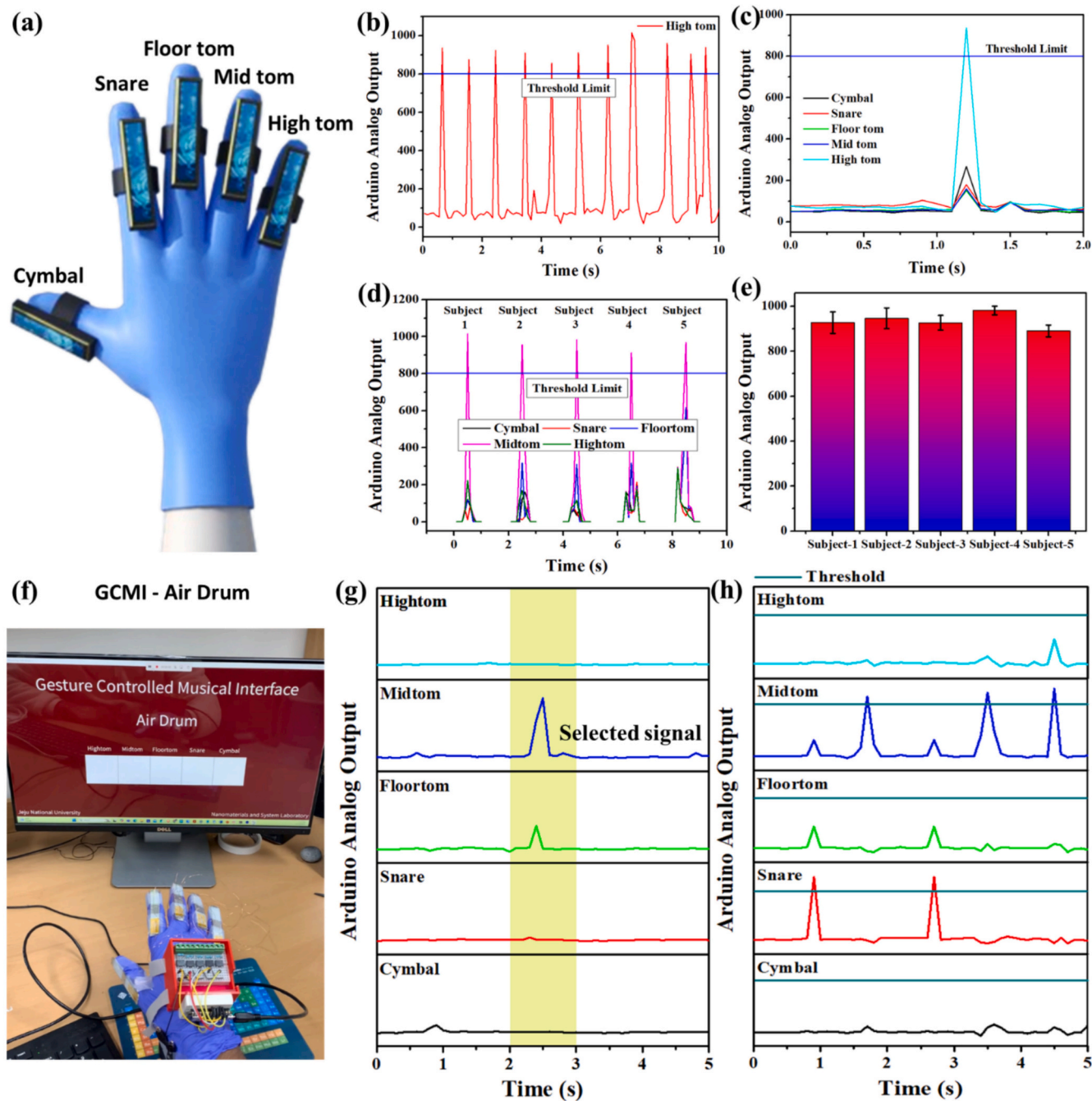


Fig. 7. (a) Scheme of GCMI smart glove, (b) single sensor response for continuous multiple flexions, (c) interference study of multiple sensors for single flexion, (d) different subject response for single instance (Mid tom), (e) error bar diagram of different subjects response for full flexions, (f) digital image of GCMI-air drum, (g) multiple sensor response of high tom selection, and (h) multiple sensor response for consecutive selection of mid tom and snare.

transducer in the smart glove for a gesture-controlled musical interface system and demonstrated real-time air drum.

4. Experimental procedure

4.1. Preparation of $0.3\text{Ba}_{0.7}\text{Ca}_{0.3}\text{TiO}_3\text{--}0.7\text{BaSn}_{0.12}\text{Ti}_{0.88}\text{O}_3$ (BCST) ceramic

BCST polycrystalline micro-particles were synthesized using the standard high-temperature solid-state reaction. High-purity analytic graded precursors like BaCO_3 (average size of 620 nm, 99% purity, Daejung Chemicals), TiO_2 (average size of 300 nm, 98% purity, Daejung

Chemicals), CaCO_3 (average size of 1 μm , 99% purity, Daejung Chemicals), and SnO_2 (average size of 120 nm, 99% purity, Daejung Chemicals) were weighed according to the stoichiometric proportions. Initially, the stoichiometric weighed precursors were homogeneously mixed in the acetone medium using the motor and pestle for 1 h. The mixed precursor powder was calcinated in a box furnace at 1350 $^\circ\text{C}$ for 6 h at a ramping rate of 5 $^\circ\text{C}/\text{min}$. The calcinated powder was ground using the motor and pestle [26]. Few drops of 5 wt% polyvinyl alcohol (PVA) solution were mixed with 2.5 g of calcinated BCST powder. The mixture was pressed into disc-shaped pellets of approximately 10 mm diameter under the applied mechanical load (10 kN to 50 kN) using the pelletizer. The pellets were sintered at 1450 $^\circ\text{C}$ for 6 h at a ramping rate

of 5 °C/min. The prepared pellets and respective grounded powders were used for the ferroelectric, piezoelectric, and electrical analysis.

4.2. Preparation of pure PDMS and PDMS/BCST composite film

The pure PDMS film was prepared using a silicone elastomer and curing agent in a 10:1 ratio (elastomer: curing agent). The homogeneously mixed polymer solution was poured into aluminium foil. Using the spin coater, the thin film was coated onto the aluminium foil at a rotation speed of 300 rpm for 2 min. The coated film was kept in the hot air oven for the polymerization process at 70 °C overnight. Similar procedures were carried out for PDMS/BCST composite film with the addition of 20 wt% BCST particles (using 30 kN load applied BCST).

4.3. Fabrication of PENG, SE-TENG, and CS-TENG

All the devices, including PENG, SE-TENG, and CS-TENG, were fabricated with dimensions of $2 \times 3 \text{ cm}^2$. For PENG device fabrication, gold was sputtered on top of the PDMS/BCST composite film to act as the electrode; on the other side, the aluminium foil served as the electrode. Copper wire was attached to both sides of the electrode. Antistatic tape was attached to both sides of the electrode to minimize external influences. PDMS/BCST-coated aluminium foils were cut to the dimensions of $2 \times 3 \text{ cm}^2$ for SE-TENG fabrication. PET sheet were attached on the back side of the aluminium foil to ensure the mechanical stability. For CS-TENG fabrication, PDMS/BCST-coated aluminium foil acts as one of the electrodes, and bare aluminium foil acts as the other electrode.

4.4. Fabrication of intrinsic hybrid nanogenerator (HNG)

For the fabrication of the intrinsic hybrid nanogenerators (HNG-1 and HNG-2), a double-layered PDMS/BCST film was coated on aluminium foil. After successfully coating the PDMS/BCST film on one side of the aluminium (which named as the Aluminium Electrode-1), copper wire was attached on the aluminium side. The PDMS/BCST second layer (which named as Aluminium Electrode-2) was coated on the aluminium foil side. Above the PDMS/BCST film, gold was sputtered and served as the electrode; a copper wire was attached to the electrode. Over the gold electrode, an adhesive PET sheet was attached. Aluminium foil was attached to the entire structure using adhesive clips, which served as the tribolayers and electrode. The copper wire is connected to the aluminium foil for the electrical measurements. With the electrical connections of the electrode, the structures were named HNG-1 and HNG-2. With the dimensions of $1 \times 6 \text{ cm}^2$, the HNG-2 electrical connection-based hybrid nanogenerator device was prepared to use as the mechano-electric sensor in the glove (termed the smart glove).

4.5. Characterization

The crystallinity and phase analysis of prepared different BCST samples were evaluated using the X-ray diffraction patterns (XRD; Rigaku; 40 kV, 40 mA, Cu-K α radiation) at room temperature. Rietveld refinement and fitting were done using the software FullProf. The symmetry and dynamic vibrations of BCST were evaluated using the Raman spectrometer (model HR Evaluation; LabRAM). Fourier transform infrared spectroscopy (FE-SEM) mapping were carried out using Nicolet iN10 MX. Using the field emission scanning electron microscopy (FE-SEM), the surface morphology and elemental analysis were carried out (TESCAN MIRA 3, Czech Republic). Using the precision material analyzer (Radiant Technologies, Inc. Precision 10 kV HVI-SC) all the ferroelectric analysis were carried out. The ferroelectric analyses for all the BCST samples were studied at a constant frequency of 10 Hz by varying the applied field from 5 kV to 40 kV. The piezoelectric coefficient for all the BCST samples was measured by using the d_{33} meter (YE2730A). Frequency-dependent dielectric studies were carried out using the impedance analyzer (HIOKI, IM3570) for all prepared BCST

samples and films by varying the frequency from 800 Hz to 4 M Hz. Piezoresponse force microscopy (PFM) measurements were carried out using the piezoresponse force microscopy (PFM) mode in Nano Scope Multimode 8 at room temperature. The mapping of surface roughness and potential of the pure PDMS and PDMS/BCST composite films was carried out using the Kelvin probe force microscopy (KPFM) mode in Nano Scope Multimode 8 at room temperature. A platinum/iridium-coated SCM-PIT-V2 probe was used for the PFM and KPFM studies. The electrical characterization includes output voltage, current density, capacitor charging, stability, and instantaneous power density were determined with the electrometer (Keithley 6514). The mechanical triggers were generated using the linear motor (LinMot, Model No: HF01–37, USA). All the electrical characterization was tested under the continuous mechanical load (a movable shaft of 2.03 Kg) with the acceleration of 1 m/s^2 to m/s^2 . The voltage divider circuit was used to measure the output voltage of the prepared devices.

CRediT authorship contribution statement

Asokan Poorani Sathya Prasanna: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Monunith Anithkumar:** Software, Formal analysis, Data curation. **Arul Saravanan Raaju Sundhar:** Software, Data curation. **Thanjan Shaji Bincy:** Methodology, Data curation. **Kwi-Il Park:** Investigation, Formal analysis. **Nagamalleswara Rao Alluri:** Writing – review & editing, Supervision, Project administration, Investigation, Formal analysis. **Sang-Jae Kim:** Writing – review & editing, Supervision, Project administration, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cej.2026.178140>.

Data availability

Data will be made available on request.

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